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GDF GLOBAL DATA FACILITY

MOBILE PHONE DATA FOR POLICY PROGRAM:

COHORT I LAUNCH WORKSHOP

LEARNING JOURNAL

WASHINGTON D.C.
OCTOBER 7-10, 2024



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0. Introduction

Welcome!

This document is provided to participants attending the Cohort 1 Launch Workshop hosted by the World Bank in Washington DC, 7-10 October 2024. It includes descriptions of the six Learning Sessions that will be delivered throughout the week, along with questions and space to write down your thoughts. During the course of this week, you will be provided with a lot of information. To help consolidate your learning along the way, we invite you to capture what you are hearing, thinking and learning in this Learning Journal. We hope that by reflecting on what is presented by the training and facilitation team, and on what you hear from your peers during group exercises and bilateral discussions, you will feel inspired, empowered and enabled to take actions that support you in your journey towards using Mobile Phone Data for Policy (MPD4P). We encourage you to capture your reflections in writing in this document. We hope that you find the sessions useful for knowledge-sharing, skills-building and action-planning. If you have any questions or doubts during the workshop, please note them down and raise them with our team either in plenary or one-to-one after the close of each day. We will do our best to support you in addressing these as we proceed through the week.

Good luck from all of us!

Co-hosts:

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Javier Burrieza, Nommon, Senior Business Development Engineer

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Sophie Delaporte, Flowminder Foundation, Communications and Marketing Manager

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I.1. An overview of pathways to success and the MPD4P Theory of Change

Session Description: This session will provide participants with an overview of the key ingredients for a successful Mobile Phone Data (MPD) initiative and describe the different elements necessary to move from the ambition of having access to MPD to the reality of using such data for policy-making. It will provide attendees with insights into the inputs, outputs, outcomes and impacts that have been considered when developing a Theory of Change for the World Bank's GDF-MPD programme in order that participants can see how the programme is expecting to bring about change in the countries receiving funding, and what support it considers will be necessary to provide.

Learning objectives: On completion of this session, participants will be able to:

- Explain what a Theory of Change is and why it can be useful for planning MPD initiatives
- Identify and organise the different elements of a Theory of Change for MPD initiatives; whereby inputs lead to outputs which lead to outcomes which lead to impact(s)
- Identify and describe some of the critical inputs that are necessary for MPD initiatives
- Identify and describe some of the expected outputs that would result from these inputs/activities
- Describe some of the outcomes that one might expect to see as a result of an MPD initiative
- Identify the key prerequisites that need to be in place for effective MPD initiatives
- Explain the value of thinking about assumptions we might be making when designing MPD projects
- Anticipate some of the pitfalls and obstacles to achieving the ambition of using MPD for policy making.

Learning Outcomes: By the end of this session, participants will have:

- More knowledge of the different components to consider when planning an MPD initiative
- An appreciation of the complexity of MPD initiatives
- An increased awareness of the need to navigate the complexity and ideas on how to do so

Reflection questions:

- Which of the key inputs for an MPD initiative do we already have in place and which will need more work?
- How confident are we in our country that we have secured what we need in order to say we have all the 'prerequisites' already in place to proceed with this initiative?
- What do I need to include in the concept note as areas for investment / research / confirmation before we move to the next step?
- What are the inputs, outputs, outcomes and impact(s) of the MPD initiative proposed for my country?

Space for reflection and action-planning:

Goals

What are the policy goals that we want to achieve by using MPD? What impact(s) do we anticipate having if we are able to access, analyse and share MPD data? Who / what are we hoping to influence, and why?

Inputs

What are the inputs we will need? Which are secured already? Which need to be planned, funded, obtained by other means?

Potential barriers

What might stop this initiative from proceeding? Are there any points of concern already identified for moving forward? Which activities or outputs are in jeopardy? What assumptions are we making which might not turn out to be true?

Other Notes

I.2 A Maturity Assessment Framework for Mobile Phone Data for Policy Initiatives

Session Description: This module provides audiences involved in the development and management of MPD for Policy initiatives with a basic understanding of the MPD Maturity Assessment Framework in order to be able to conduct initial assessments of their respective initiatives. This module complements the Theory of Change module as a further tool to identify how to achieve impact with an MPD for Policy initiative.

Learning objectives: At the end of this module, participants will be able to:

- Explain why they would use the Maturity Assessment Framework in their initiative
- Describe the key characteristics of the different maturity stages
- Identify the three overarching themes and eight core dimensions of an MPD for Policy initiative
- Apply the maturity framework to their own MPD for Policy initiative, at a high-level and within the scope of the available information.

Learning Outcomes: By the end of this module participants will have:

- An understanding of the purpose and value of the Maturity Assessment Framework
- An understanding of the broad range of dimensions that contribute to an MPD for Policy initiative
- Confidence to guide other stakeholders through an initial maturity assessment of their MPD for Policy initiative
- Skills to conduct an initial maturity assessment of their MPD for Policy initiative

Reflection questions:

- What are the areas in which your MPD for Policy initiative is strongest?
- Which dimensions pose the greatest barriers or risks to your MPD for Policy initiative?
- For which dimensions do you have a strong understanding of the maturity of your MPD for Policy initiative?
- Which stakeholders do you need to engage to solicit the necessary information to complete a maturity assessment of your MPD for Policy initiative?

Space for reflection and action-planning:

Maturity Assessment

Recalling the different dimensions of maturity (see table below), which of the dimensions are well developed in our context? Which are less mature and need some focused attention? How confident do we feel that we can accurately assess each dimension for our context? What are the dimensions in which the MPD4P program could provide support, why and how?

Tier 0	Tier 1
Feasibility	SUITABILITY OF LEGAL AND GOVERNANCE ENVIRONMENT
	STAKEHOLDER ENGAGEMENT AND ALIGNMENT
	SUITABILITY OF DATA INFRASTRUCTURE
	SUITABILITY OF DATA ENVIRONMENT
	HUMAN RESOURCE CAPACITY AND AVAILABILITY
Impactfulness	IDENTIFICATION AND IMPLEMENTATION OF APPLICATIONS
	POTENTIAL FOR AND PROTECTIONS AGAINST MISUSE
Sustainability	SUSTAINABILITY PROSPECTS

Stakeholder engagement

Are we missing information or making assumptions about what stages we are at, per dimension? Who do we need to engage to help us answer questions we may have about the level of maturity for each dimension in our context? Who could help us to fill in the gaps in our knowledge or understanding of how ready we are to use MPD for Policy? If we feel ready, what more can we do to strengthen our maturity across the framework?

Other Notes

2.1 Data Characteristics and Quality Aspects

Session Description: This session will cover the following topics:

- Introduction to CDR data
- Key information included in CDR Data
- Strengths and limitations of CDR Data
- Quality considerations in using CDR data for policy purposes
- Adjusting biases in statistics derived from CDR Data
- Quality assurance framework

Learning objectives: On completion of this module learners will be able to:

- Describe what CDR data is and how it is generated
- Describe key information included in CDR data
- List some of the strengths and limitations of CDR data
- Describe quality considerations for using CDR data for policy purposes
- Describe the data or surveys for adjusting biases, how this can be done, and some of the pitfalls or challenges involved

Learning Outcomes: By the end of this module participants will have:

- Knowledge of the key information included in CDR data and how they are generated
- Knowledge of the key strengths and limitations of CDR data
- Knowledge of quality consideration in using CDR Data for policy purposes
- Confidence in using CDR data for statistical data production, with knowledge of how to ensure quality
- Knowledge of data/surveys and methods for adjusting biases
- Confidence in monitoring and improving statistics produced from CDR data

Reflection questions:

- What have I learnt about the key information in CDR data and how this information is generated? What implications does this have for our plans in my context?
- How will the strengths and limitations of CDR data affect how I want to use it in the future?
- How am I going to ensure that the data we will use is good quality? What quality factors will we need to consider when we are interpreting any statistics derived from the data?
- What practical steps could we take in my context to mitigate the limitation of CDR data? Are there any existing sources of data that could be used for adjusting biases? Are there opportunities coming up for collecting such data?

Space for reflection and action-planning:

Data sets

What data are we requesting, and from which organisation(s)? Are we clear about the specifications of the data required? Do we know what variables we'll be analysing? Have we communicated our dataset specifications and variables to the data provider(s) clearly? Do we understand what format the data is held in and what that means for analysis? What concerns might we have about data availability, at this time? What else do we anticipate being a concern?

Data quality

How confident are we that we will get good quality data? What can we do to ensure that good quality data is received, or when it's not that we can fix the situation? What checks could we put in place to ensure the data is quality assured? Who will do that quality assurance?

Data biases

What biases will the data we use probably have? What can we do to address these biases and adjust for them? What other data sources might we want to look at for adjusting the bias of the data we use? What methods would we be using? Who will do this adjustment work? How important will it be to adjust for bias, in our context and for our use case?

Other Notes

2.2 Data Processing, Privacy and Pipelines

Session Description: This session provides representatives from Cohort 1 countries with intermediate-level skills and knowledge needed to understand the steps for transforming raw MPD into useful statistics. The module covers key areas such as data ingestion, processing, aggregation, data enrichment techniques, privacy preservation, and statistical disclosure control.

Learning objectives: On completion of this session, participants will be able to:

- Identify the fundamental components of a data pipeline for MPD, including data sources, data cleaning, data processing, and output stages
- Describe data enrichment techniques for MPD, such as identifying home and work locations from spatio-temporal patterns based on frequency, duration, and timing of visits
- Explain the various data sharing models, including centralised and federated approaches, and their implications for collaboration
- Explain the importance of privacy in MPD, including the legal and ethical considerations for data collection, processing, and dissemination
- Describe MPD aggregation methods, including statistical disclosure control (SDC) techniques, and explore best practices for effective data dissemination

Learning Outcomes: By the end of this module participants will be have:

- Knowledge about the architecture of a complete data pipeline for MPD, including the key components and their functions, from data acquisition to dissemination
- Knowledge of the privacy risks associated with MPD
- Knowledge about steps to apply privacy-preserving methods that comply with legal and ethical standards, such as GDPR and fundamental principles of official statistics

Reflection questions:

- What are the main components of an MPD pipeline, and how do they work together?
- What are the differences between data sharing models, and which is most suitable for my use case while maintaining privacy and security and sustainable statistical production in the long run?
- What are the best practices for aggregating MPD for dissemination purposes?

Space for reflection and action-planning:

Data access

What approach are we taking to obtaining the data we need? Who will receive or be able to access what data, at what level of granularity/detail (raw, pseudonymised, aggregated)? What data does our use case need as input data? What frequency and level of spatial & temporal resolution are we expecting as part of the data flow?

Data processing

What are the data processing steps that will/should occur at different stages for a specific use case? How is the data stored and how will it be transmitted between parties securely? What processes should be taken for cleaning the data and who will be responsible for it? Who will do the processing? How should the data be aggregated for a specific use case?

Data privacy and protection

What safeguards will we put in place to ensure strong data protection and data privacy in different stages of the data pipeline? What concerns might different stakeholders have? Besides privacy, which other ethical concerns should be taken into account when dealing with MPD? What standards for data protection and privacy will we adopt? What risks will we face if the standards are not met?

Other Notes

2.3 Institutional arrangements: Partnership, legal, regulatory and financial considerations

Session Description: This session provides participants with guidance on the following topics:

- Data Access Models and good practices
- Financing/regulatory models - short and longer term
- Setting up of MOUs/Legal agreements

After the session, participants will be able to identify important concepts and choose between possible scenarios to set up the necessary legal agreements for data acquisition.

Learning objectives:

The objective is to help participants identify the relevant topics and examples on which to build their understanding of possible institutional arrangements. The session isn't exhaustive; but provides different examples of countries choosing various institutional arrangements that are especially important for learning. By the end of this session, participants will:

- Be able to determine the roles of each stakeholder in data governance for setting up agreements.
- Know key enablers and risks in a country's legal and regulatory framework.
- Understand various data acquisition models and how these influence partnerships with MNOs.
- Be equipped to identify the most relevant funding mechanism and financial sustainability options for the project.
- Be able to determine key points to include in a Data Sharing Agreement before drafting.

Learning Outcomes: By the end of this module participants will have:

- Knowledge on how to identify benefits and costs for MNOs to participate in the project
- Knowledge on different data sharing models
- Knowledge on key points of a Data Sharing Agreement that need agreement between parties

Reflection questions:

- What role does each stakeholder play in the initiative?
- What are the key enablers and risks in my country's statistical and telecommunications legal and regulatory environment?
- Which data acquisition models are options for the initiative?
- How to determine the key points of a Data Sharing Agreement that need agreement between parties before drafts can be generated?
- What is the most relevant funding mechanism for the project?

Space for reflection and action-planning:

Data sharing incentives

What are the motivations of the data provider(s) / data controller(s) for sharing data? What incentives exist in the data ecosystem for enabling data sharing? Which of these incentives are options for us to use in our initiative? How will the incentives for sharing data be sustained over time or change?

Data acquisition

Have we determined the institutional arrangement for how data will flow between parties?
Can we capture the data flow of institutional arrangements in a diagram?

What steps do we still need to take, if any, to sort out the partnering arrangements between the different stakeholders? Do we have a relevant written agreement in place which signs the parties up to these arrangements? What steps are we taking to enable the institutional arrangements to be sustained in the long-term?

Data sharing agreements

What are the data governance arrangements for our proposed initiative? What topics should we cover in any data sharing or partnership agreement documents? Which areas of such an agreement are already known and clear, which areas are less certain? Do we have a clear table of contents or structure in mind for such an agreement? Where can we go for some good examples or who can we learn from?

Financial considerations

Do we know all of the costs that are going to be associated with this initiative? Do we have a clear budget available for covering the costs associated with all the steps including data acquisition and storage, data processing, data dissemination, etc? Which costs are we assuming are already covered and are they really going to be available for the work? Which staffing are we hoping will be provided through in-kind contributions? What infrastructure, hardware, capacity building or software costs need to be added and how are we covering these? What will happen if unforeseen costs arise, is there any contingency to cover such costs somehow?

Other Notes

3.1 MPD Policy Applications

Session Description: This session will provide participants with an overview of policy applications of MPD, illustrating them with real-world examples from previous and ongoing case studies. Based on well-established international guidelines on the application of MPD for policy-making, the session will cover a variety of cases with different levels of complexity and impact. Participants will be informed of the key requirements for MPD processing for each application and shown examples of output indicators for each application.

Learning objectives: on completion of this module learners will be able to:

- List at least two sources of guidance on how MPD can be used for different applications
- Identify at least one type of final user besides NSOs that may benefit from MPD-based information in each application
- Describe the expected complexity of a given MPD application and identify the required pre-conditions for implementation (i.e. intricacy of the MPD processing algorithms, data fusion requirements, acceptable precision in the output indicators)
- Describe the expected impact on decision-making of a given MPD application
- Describe at least one example of how MPD has been used in a practical/real-world application
- Identify at least one policy application for a given MPD-derived indicator
- List at least two lessons learnt from MPD policy applications

Learning Outcomes: by the end of this module participants will have:

- Knowledge of the state-of-the-art references for the use of MPD in official statistics
- Skills to pre-assess the complexity and impact of each potential MPD policy application, as well as to estimate the required resources to implement an MPD initiative
- Confidence to explain the benefits for NSOs and other potential final users of the use of MPD
- Skills to specify output MPD-derived indicators that are useful for several policy applications

Reflection questions:

- Which resources can I use to find information about potential MPD policy applications?
- Who are the key stakeholders to involve in each application domain?
- Who are the final users who can benefit from MPD-derived information in each application

domain?

- What information do I need to retrieve in my country to assess the complexity and impact associated with a given MPD policy application?
- How can any of the early applications of MPD in my country, if any, be mapped to the application domains covered in the session?

Space for reflection and action-planning:

Priority applications

What are the priority use cases that we should be pursuing, taking into account what we've learnt this week about MPD and using MPD for policy? Which use cases do we think might be most impactful in our context? Which uses would we love to do but we're unsure how to do them right now?

Developing applications

Which organisations or individuals do we need to engage in order to pursue the planned or priority use cases? Are they already engaged? If not, how will they be incentivised or enabled to engage with this initiative? What data do we think others will be able to provide for use cases / applications?

Communication and dissemination

How will initiative and the results from the analysis and interpretation of MPD be shared with relevant stakeholders? How will policymakers be informed of the use of MPD-derived indicators or statistics in order to secure their buy-in and trust in the data? How will messaging about this initiative be done to secure public support or prevent a negative backlash from misunderstandings or objections?

Other Notes

MPD4P Country-level Action Plan Summary - Template

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[Country Name MPD Action Plan]

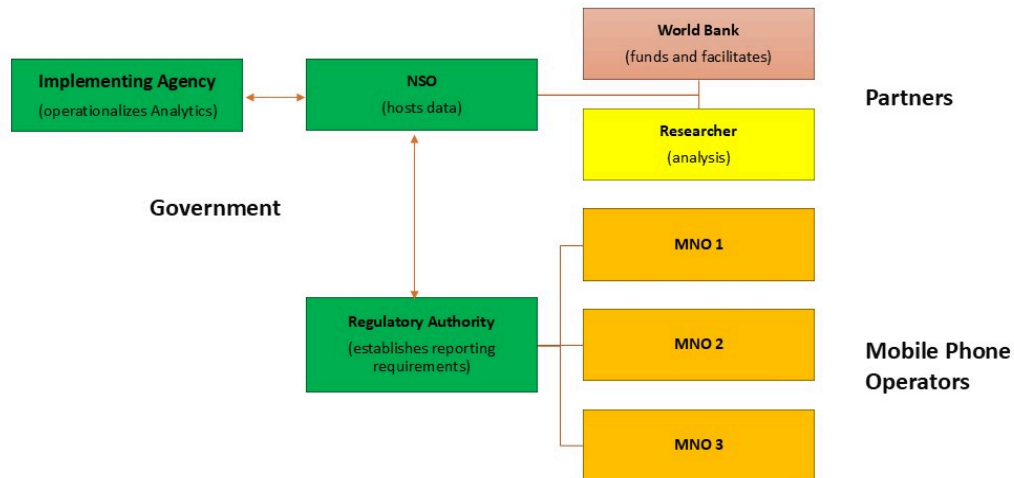
Executive Summary

Instructions: This will be the slide each team presents on Thursday. Please draw from the worksheets to complete. Presentations should be 3-5 minutes

Objective	Level of Maturity	Key Stakeholders	Use-Case	Top 3 Challenges
i.e. To leverage [MNOs] data to inform XX	- Pre-foundation - Foundation - Practitioner - Expert	- MNO - Regulator - NSO - Implementing agency - Other	I.e. Mobility, Poverty Mapping, Targeting	I.e data infrastructure, data security/confidentiality, capacity building

Stakeholder Mapping

Instructions: Map out the various stakeholders, their respective responsibilities and the flow of data between them [see example below]



NEXT STEPS

Instructions: draw from the detailed action plan to list the top three actions you intend to take

Action 1	• Description: • Responsible Party: • Timeline:
Action 2	• Description: • Responsible Party: • Timeline:
Action 3	• Description: • Responsible Party: • Timeline:

THANK YOU - [COUNTRY TEAM]



[take a team picture and post it here!]



5



ANNEX





The GDF-MPD window is a multi-partner window funded by the Spanish Government, Ministry of Economy, Commerce and Business. The GDF-MPD program is jointly managed by DEC (DECDG, DIME), Poverty and Digital Development Global Practices in technical partnership with the International Telecommunications Union (ITU).

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